Chain rule

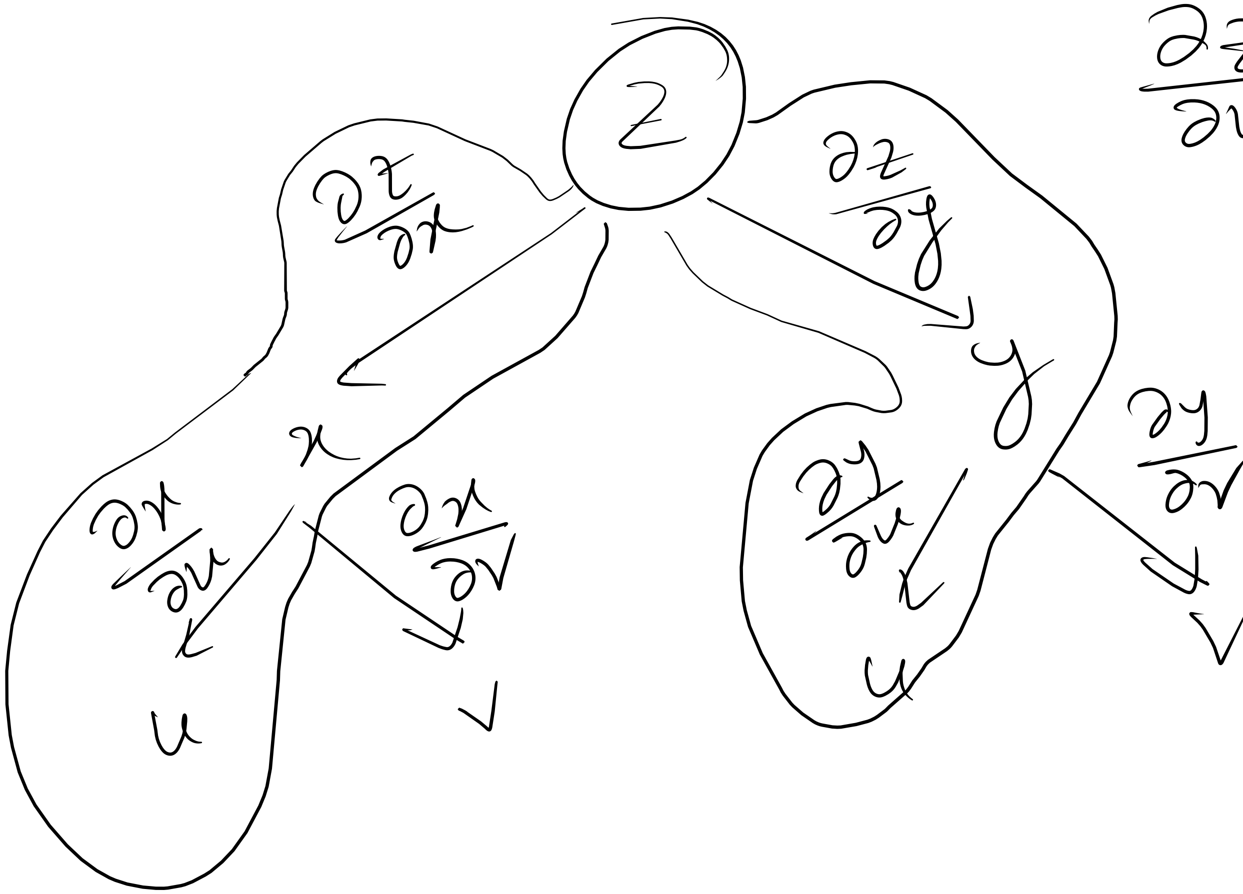
$$\begin{tikzcd}
 & z \arrow[dl, "\frac{\partial z}{\partial x}"'] \arrow[dr, "\frac{\partial z}{\partial y}"] & \\
x \arrow[d, "\frac{dx}{dt}"'] & & y \arrow[d, "\frac{dy}{dt}"'] \\
t & & t
\end{tikzcd}
\qquad
\begin{tikzcd}
 & w \arrow[dl, "\frac{\partial w}{\partial x}"'] \arrow[d, "\frac{\partial w}{\partial y}"'] \arrow[dr, "\frac{\partial w}{\partial z}"] & \\
x \arrow[d, "\frac{dx}{dt}"'] & y \arrow[d, "\frac{dy}{dt}"'] & z \arrow[d, "\frac{dz}{dt}"'] \\
t & t & t
\end{tikzcd}$$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\frac{dx}{dt} + \frac{\partial z}{\partial y}\frac{dy}{dt}$$

$$\frac{dw}{dt} = \frac{\partial w}{\partial x}\frac{dx}{dt} + \frac{\partial w}{\partial y}\frac{dy}{dt} + \frac{\partial w}{\partial z}\frac{dz}{dt}.$$

If $x = x(u,v)$, $y = y(u,v)$

$$z = f(x,y) = f(x(u,v),\, y(u,v)).$$

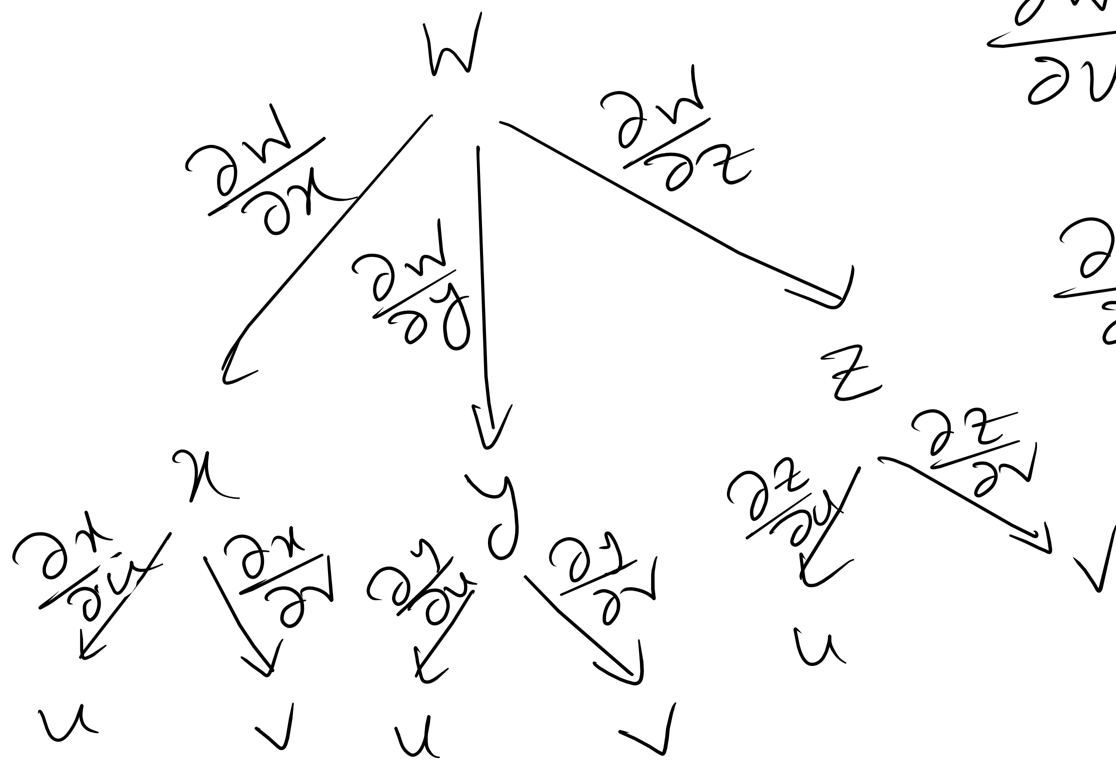


$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$$

$$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial v} + \frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial v}$$

If $x = x(u,v)$, $y = (u,v)$, $z = (u,v)$

$$W = f(x,y,z) = f(x(u,v), y(u,v), z(u,v)).$$



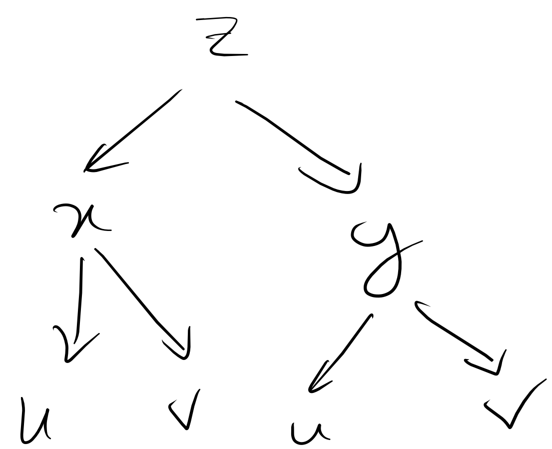
$$\frac{\partial w}{\partial u} = \frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial u} + \frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial u} + \frac{\partial w}{\partial z}\cdot\frac{\partial z}{\partial u}$$

$$\frac{\partial w}{\partial v} = ??$$

H. W.

Ex: $z=f(x,y)$, $f(x,y)=e^{xy}$, $x(u,v)=3u\sin v$, $y(u,v)=4uv^2$

z_u, z_v,



$$z_u=\frac{\partial z}{\partial x}\cdot\frac{\partial x}{\partial u}+\frac{\partial z}{\partial y}\cdot\frac{\partial y}{\partial u}$$

$$=e^{xy}\cdot y\cdot 3\sin v+x e^{xy}\cdot 4v^2$$

$$=4e^{12u^2v^2\sin v}\,uv^2\cdot 3\sin v+3u\sin v\,e^{12u^2v^2\sin v}\,4v^2\cdot$$

$$=12uv^2 e^{12u^2v^2\sin v}\sin v+12uv^2\sin v\,e^{12u^2v^2\sin v}$$

$$=24uv^2\sin v\,e^{12u^2v^2\sin v}$$

* For a differentiable function $f(x,y)$ with continuous second partial derivatives $x = r\cos\theta$, $y = r\sin\theta$. Show that

(i) $f_r = f_x \cos\theta + f_y \sin\theta$.

(ii) $f_\theta = -r f_x \sin\theta + r f_y \cos\theta$

(iii) $f_{rr} = f_{xx} \cos^2\theta + 2 f_{xy} \cos\theta \sin\theta + f_{yy} \sin^2\theta$

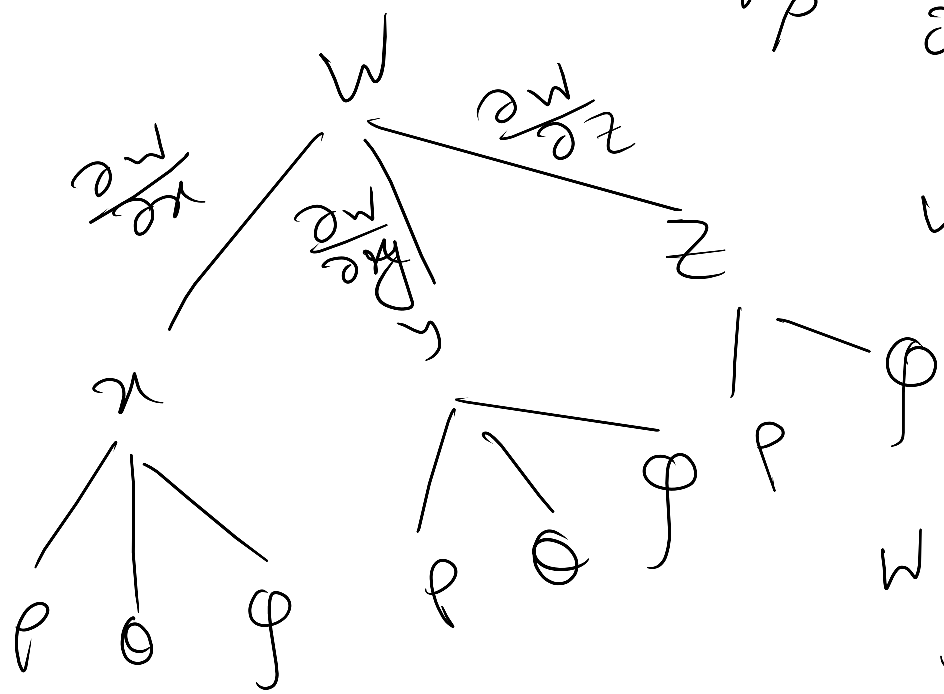
(iv) $f_{\theta\theta} = r^2 f_{xx} \sin^2\theta - r^2 f_{xy} \sin 2\theta + r^2 f_{yy} \cos^2\theta$
$- r f_x \cos\theta - r f_y \sin\theta$.

* $w=\sqrt{x^2+y^2+z^2}$, $x=\rho\sin\varphi\cos\theta$, $y=\rho\sin\varphi\sin\theta$, $z=\rho\cos\varphi$

$x=r\cos\theta$, $y=r\sin\theta$, $z=z$, w_r, w_θ, w_z -

$w_\rho, w_\theta, w_\varphi = ??$

$$w_\rho=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial \rho}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial \rho}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial \rho}$$



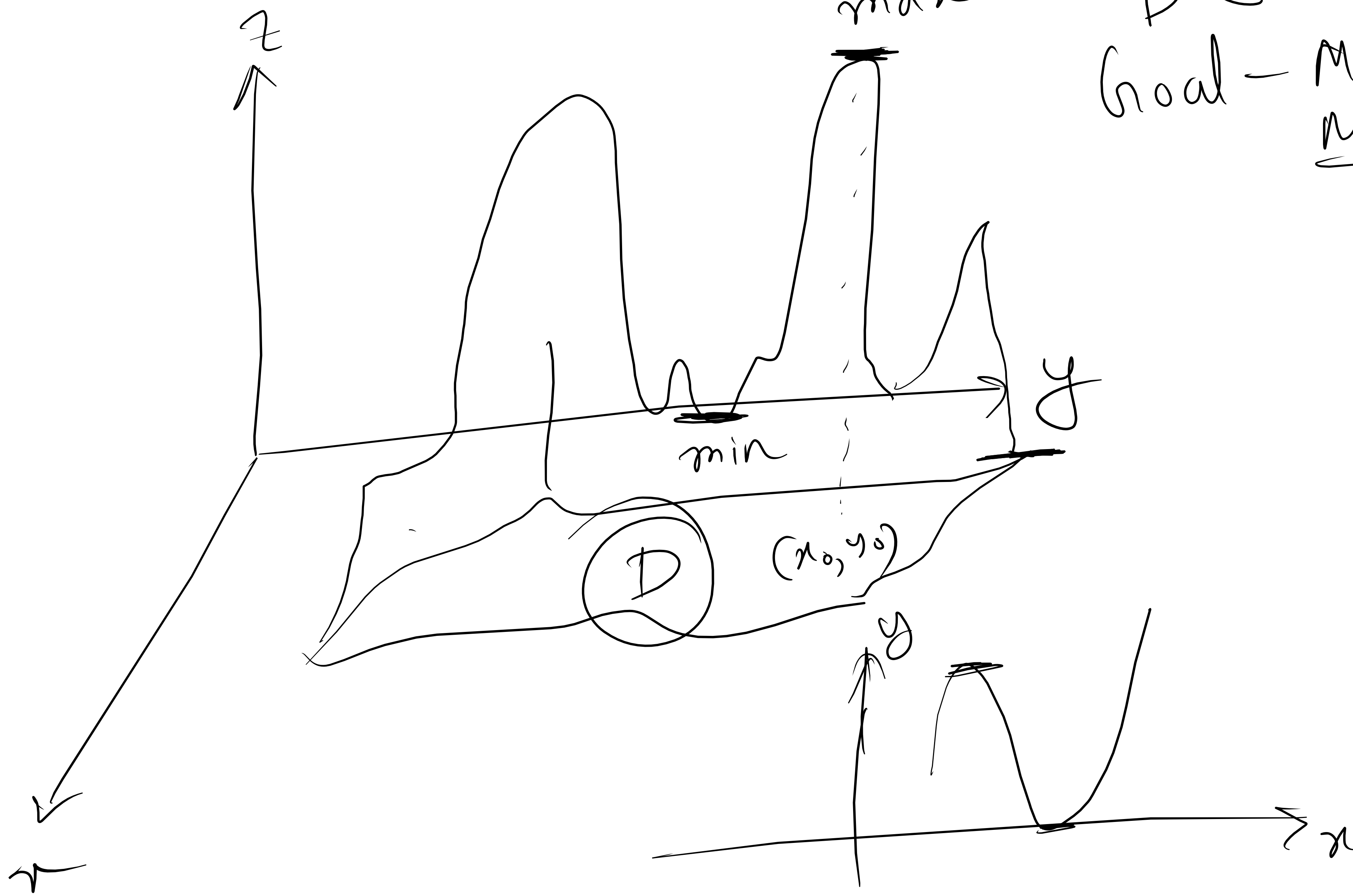
$$w_\theta=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial \theta}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial \theta}$$

$$w_\varphi=\frac{\partial w}{\partial x}\cdot\frac{\partial x}{\partial \varphi}+\frac{\partial w}{\partial y}\cdot\frac{\partial y}{\partial \varphi}+\frac{\partial w}{\partial z}\frac{\partial z}{\partial \varphi}$$

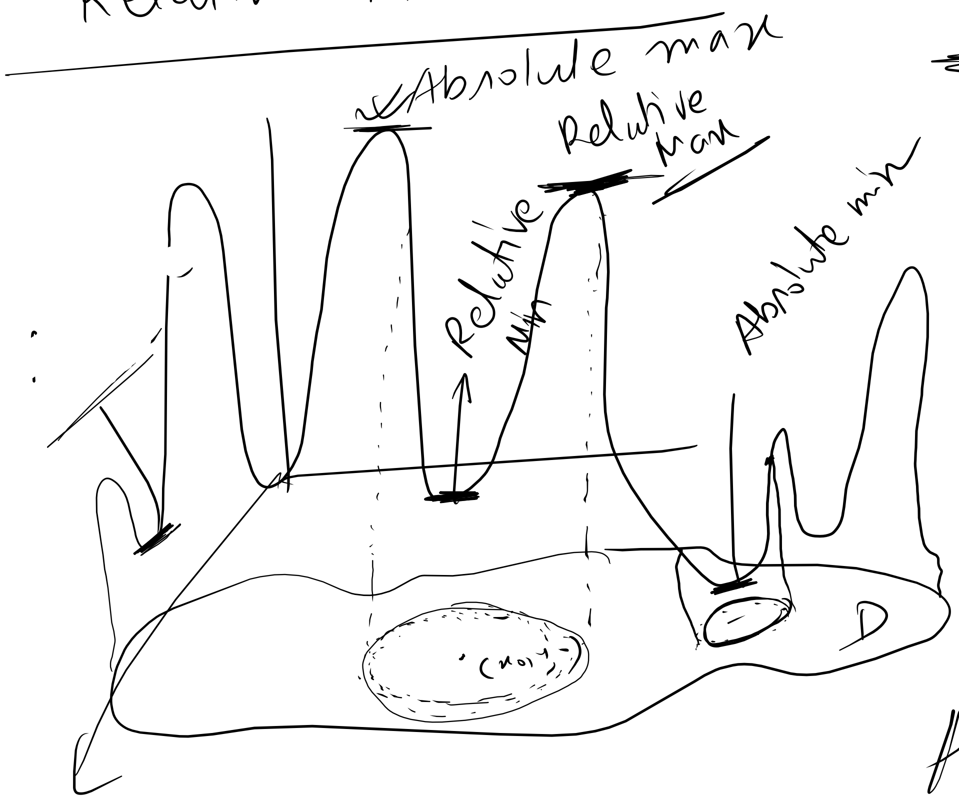
Maximum/minimum for two variables:

$D \subset \mathbb{R}^2 \to (x, y)$

Goal - Max
min



Relative Maximum (Min)



$f(x,y) \to$ relative maximum (min)

at $(x_0, y_0) \to$ center open disk

such $f(x_0,y_0) \geq f(x,y) \to$ max $\forall (x,y)$ in the

$f(x_0,y_0) \leq f(x,y)) \to$ min

open disk

Absolute Maximum (Min)

$f(x,y) \to$ absolute Maximum (Min)

at (x_0, y_0) such that $f(x_0,y_0) \geq f(x,y) \to$ max

$(f(x_0,y_0) \leq f(x,y)) \to$ min

$\forall (x,y)$ in the set

D.

Section-13.8

Extremum (Maximum / minimum).

Critical points: A point (x_0, y_0) in the domain of a function $f(x,y)$ is called a critical point of the function if $f_x(x_0, y_0) = 0 = f_y(x_0, y_0)$ / one or both partial derivatives do not exist at (x_0, y_0).

Single Variables $\left[f'(x_0) = 0 \;/\; \text{function is not defined} \;/\; f'(x_0) \text{ does not exist} \right]$

Saddle point: Saddle point is a critical point which is not a local extremum (relative) of the function.

H.W. partial Derivative & Chain rule + Outline

H.W. sheet